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CPE 301

3/19/2026

Final Project Part 2

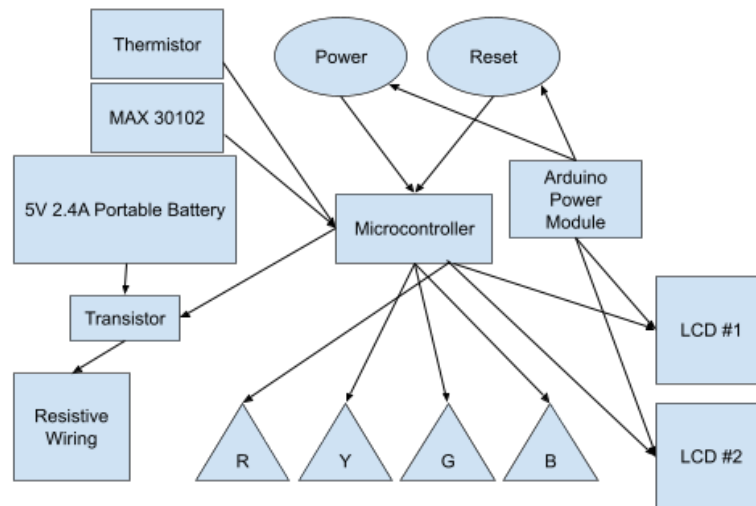
1. Project Introduction

Medical technologies, devices, and care have improved significantly over the past decades due to advances, optimizations, and new research in many different medical fields. Despite medical advances, the proportion of newborn deaths among all child deaths is increasing (Lunze, 1). This points to a global lack of technological advances in neonatal care in comparison to other medical fields. According to the World Health Organization, 78% of these deaths occur in developing countries (WHO). In fact, there is a statistically significant relationship between neonatal mortality rates and a country's median income as seen in *'Neonatal Admission Temperature in Middle- and High-Income Countries'* (Garcia). Clearly there is a need for cheap, accessible, and life-saving neonatal devices, especially for lower income countries who lack medical technologies, trained staff and money to fix these issues. And while there is no tested statistical relationship in my sources between lower income countries and non-lower income countries due to a lack of existing data and ability to create accurate statistical studies; It is a reasonable assumption that this relationship would only continue, or even show an exponentially greater increase in neonatal death percentages as income decreases. If this were the case, it would highlight the importance of new cheap, accessible, and potential life-saving neonatal devices. Among these deaths, the leading causes are premature birth, asphyxia and infections with a substantial amount of deaths being in comorbidity with hypothermia (WHO and Lunze, 1). Hypothermic babies in middle and higher income countries are shown to have an almost double lower chance of survival than eutermic babies (Garcia). The proposed embedded system would be a part of a larger affordable neonatal swaddle kit that monitors breathing, body temperature, and contains an alert system as well as basic treatments for asphyxia and hypothermia. It would be targeted to hospitals, or homes because at-home births are common in low income countries, specifically in South Asia and Sub-Saharan Africa, which is where 78% of all neonatal deaths occur

(WHO). This entire swaddle system is meant to help detect and combat hypothermia and asphyxia. The specific embedded system I am creating / writing about for CPE 301's Final Project is the axillary temperature detection system used to detect and alert that the baby body temperature has dropped beneath 36.0 degrees celsius, the highest temperature classified as moderate hypothermia. For this device component, an embedded system is the most reasonable solution because it is a smaller computer system that performs a specific task within the larger swaddle system.

2. Proposed System

a. System Schematic Diagram



b. Component List

- Arduino Mega 2560
- Arduino Power Supply Module
- 5V 2.4A Portable Battery
- LCD 1602 Modules (Two)
- LED (Four: Green, Yellow, Red, and Blue)
- Button (Two)

- Thermistor
- MAX 30102 Sensor
- NPN Transistor PN2222
- Resistors (4 330 Ω , 2 4.7 k Ω , 2 10k Ω)
- Electrical Wires
- Resistive Wiring
- TPU Film
- Nylon Repair Tape
- Swaddle

3. System Behavior

a. Module Design

The first module my device consists of is the control module. It consists of a Arduino Mega 2560 and two external power sources. The control module is responsible for managing and powering all device components. It also receives and integrates data from the sensing module, receives input from the user interface module, controls the actuating output of the heating module via PWM, and controls the display module.

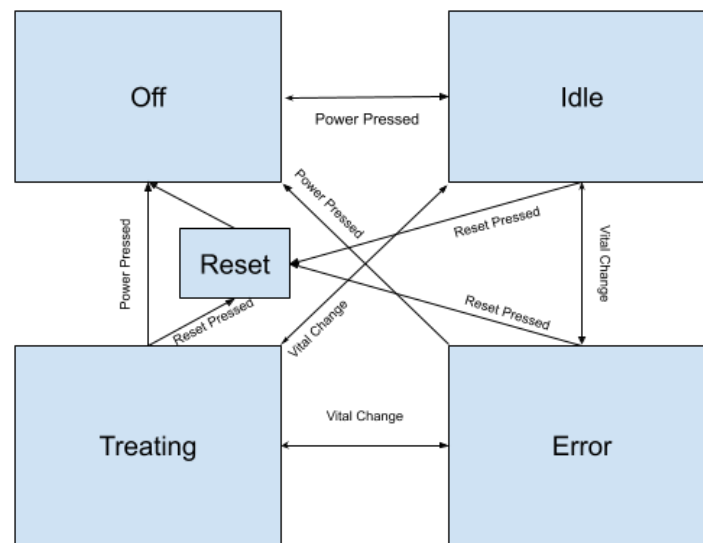
The second module, mentioned above, is the sensing module. It consists of a MAX 30102 chip, capable of measuring a patient's pulse and blood oxygen content, and a thermistor which measures the user's body temperature. It receives directions from the control module on how often to measure data, and transmits the requested information back to the control module for integration.

Next, the user interface module consists of two buttons which are used to navigate the device's state and provide a safe reset. The power button changes the device state to 'off', or from 'off' to 'idle'. The reset button changes clears all data and any current treatment and returns the device to its 'off' state. The user interface module receives power and transmits data to the control module.

The heating module consists of resistive wiring, a TPU bilayer, nylon repair tape, and a standard neonatal swaddle. The heating module safely transfers electrical energy into thermal energy via a safe, soldered connection from a jumper wire to resistive wiring. The resistive wiring is snaked across and evenly heats the TPU bilayer which is attachable to a child safe fabric swaddle. It receives power and commands from the control module.

Lastly, the display module is responsible for 1) constantly displaying the patient's vitals and 2) displaying the correct treatment to a second LCD. The display module also consists of four LED's which is used to signal the system's state. It receives commands and power from the control module.

b. Control Logic (State Diagram)

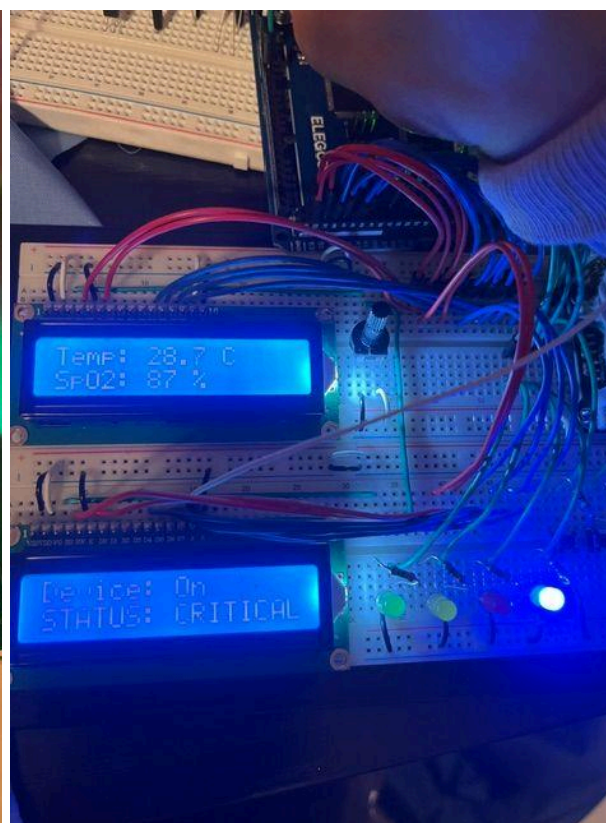
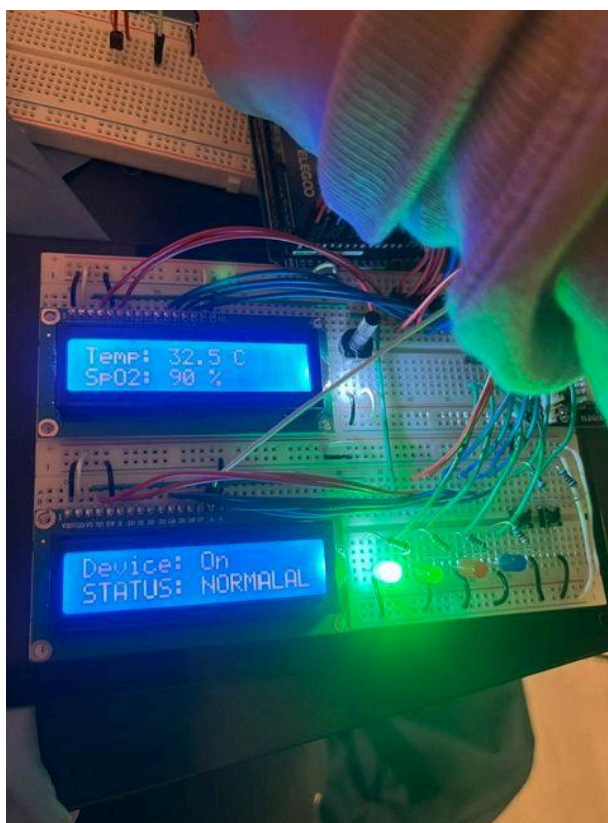
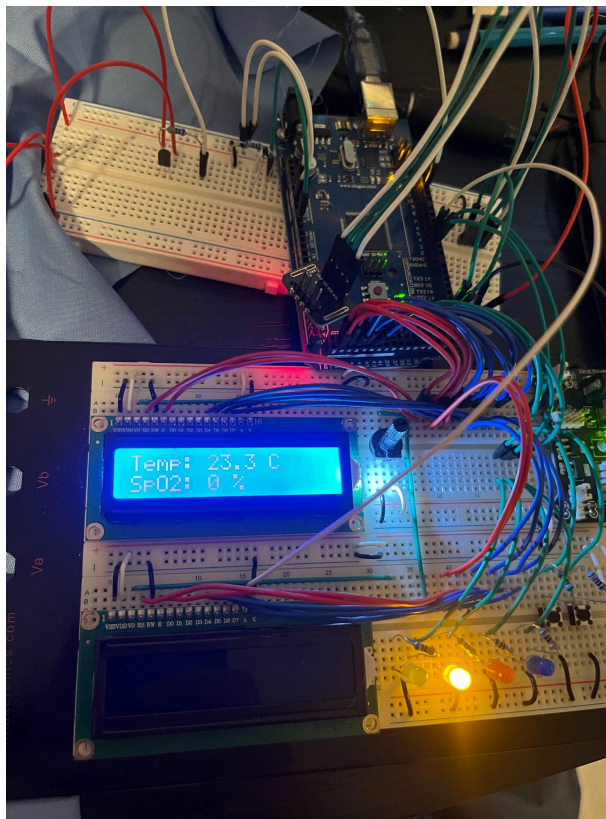
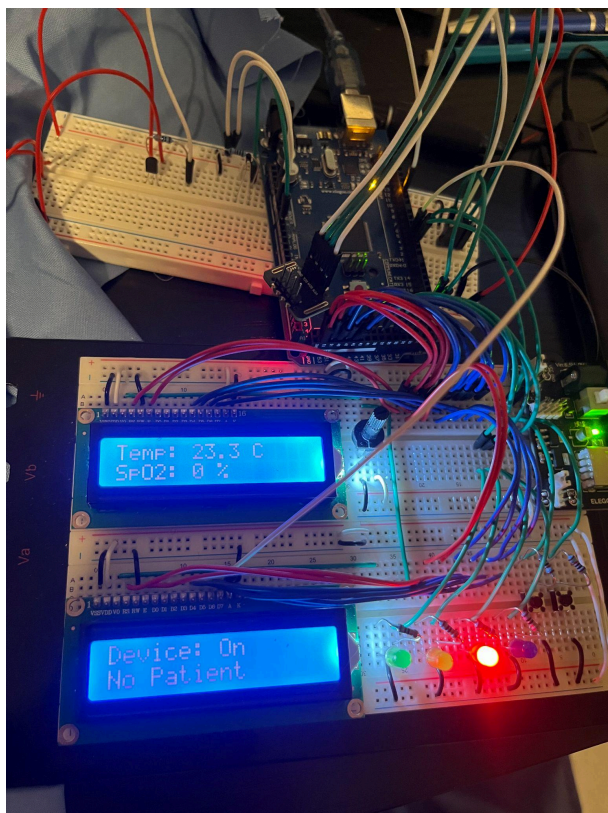


1. **OFF:** The system is in standby mode. The device is connected to charged power source but the LCD, LED's, Actuators, and Temperature Sensors are all inactive. The only running electricity is from the device's small yellow LED that signals it is currently lit up. This state is triggered by pressing the off button during any other state and can only be exited by pressing the device's on button.

2. IDLE: The device monitors the neonatal's temperature and lights up the green LED. The idle LED is on and the temperature sensors take measurements that are recorded and displayed on the LCD display. The state is triggered by pressing the on button while off or pressing the reset button while in the error state. The device can leave this state by the off being pressed, which returns the device to its non-monitoring mode. It can leave this state by detecting an error and entering the error state. Or, the device will leave this state when it senses one of the neonate's vitals are in critical condition, causing it to enter its treating state. The device can also leave this state by the reset being pressed, which activates the reset ISR.
3. Treating: In the on state, the device still monitors the neonates temperature, signals something is wrong with a red LED and also decides a "treatment" based on the neonate's temperature (either heating or cooling). It can exit the on state in four ways: 1) the off button being pressed, 2) the babies' temperature returns to a normal value, or 3) the caretaker presses the reset button, or 4) the vitals no longer are readable and it enters the error state.
4. ERROR: If the device is powered on, and vitals are not able to be correctly read. The device enters its error state and signals to the user that something needs to be adjusted.

4. Result

a. Circuit Images




```
21:47:21.860 -> [14:00:00] STATE: OFF | Temp: 23.3C | SpO2: 90.2%
21:48:22.159 -> [34:30:00] STATE: NORMAL | Temp: 35.7C | SpO2: 89.0%
21:49:22.501 -> [34:39:00] STATE: HEATING | Temp: 32.3C | SpO2: 81.8%
21:50:22.822 -> [34:31:00] STATE: ALARM | Temp: 23.3C | SpO2: 0.0%
```

5. Environmental and Energy Impact

The major motivation for this design is to help reduce neonatal deaths in lower income countries. In this places, lack of detection and treatment technology often to leads to late, efficient, or no medical intervention. This embedded system aims to provide early detection and prevention, while also providing basic treatment for neonates suffering from hypothermia that does not require Kangaroo Care. To ensure that our system is affordable and accessible, the system should also be designed and distributed as cheaply as possible by using cheap components or even recycled parts and materials.

The system's state based design allows for maximum efficiency by minimizing the swaddle's power consumption while there is no need to intervention. The device includes a hard reset and hard shutdown function which helps to ensure safety and a safe error recovery process if any portion of the device were to fail. Lastly, the device is aimed to be extremely user friendly and not require extensive medical training for healthcare providers in non-developed settings, common issues in areas with lower quality healthcare.

6. Conclusion

- a. Yes the device and code worked. However we are limited to the course's arduino kit, so this version can at best, only be a prototype. The use of the arduino based setup also mandates constant connection to a computer and would be infeasible for a swaddle technology. The technology if also restricted to components that are found in our given arduino kits. Therefore, this embedded system is at best a proof of concept which requires significant development and improvements before becoming a real world practical solution.

Final Submission

- Report (pdf)
- Github:
 - https://github.com/stevenjuan1799-arch/CPE301_FinalProject.git
- Video of Operation
 - Github does not allow for large files, so I am also attaching here in case it does not allow.
 - [6847](#)